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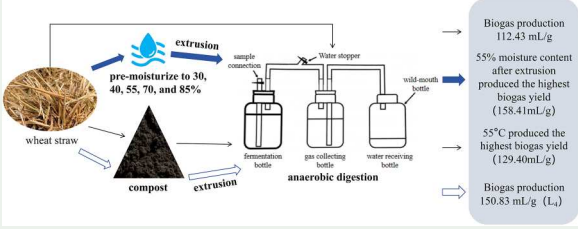
The effects and underlying mechanisms of composting and extrusion pretreatment on anaerobic biogas production from wheat straw

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ABSTRACT

The effects of moisture + extrusion and composting + extrusion treatments on the production of biogas from the anaerobic digestion (AD) of wheat straw (WS) were systematically investigated. The results showed that when the moisture content of WS was adjusted to 55% before extrusion and maintained for 24 h, the biogas production of both the unextruded and extruded samples reached the highest levels, which were 130.48 and 158.41 mL/g, respectively. Based on these results, the WS was further aerobically composted after adjusting its moisture content to 55%, and then extruded for biogas production. The biogas production of unextruded WS initially increased and then decreased, reaching the maximum of 129.40 mL/g when the composting temperature rose to 55°C. In contrast, the biogas production of extruded WS increased at all composting temperatures and reached the maximum of 150.83 mL/g at 55°C, which is 19.49% higher than the initial biogas production of extruded WS at the start of composting. Mechanistic analyses indicated that extrusion reduced particle size, increased specific surface area, and disrupted lignocellulosic structure, while composting promoted partial degradation of hemicellulose and lignin. These structural modifications enhanced substrate accessibility and microbial degradation, thereby boosting biogas production. In conclusion, the combination of composting and extrusion pretreatment can significantly improve the biogas production performance of anaerobic digestion of WS, with an increase ranging from 20.76% to 43.45%, which is worthy of further research.



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KEYWORDS
Composting pretreatment; extrusion pretreatment; biogas; wheat straw; anaerobic digestion

1. Introduction

In recent years, there has been a growing focus on the environmental pollution caused by agricultural waste. China produces approximately 900 million tons of crop straw annually [1]. Rice straw, maize stalks, and wheat straw constitute the primary categories of straw, accounting for about 90% of the global total [2]. The treatment, recovery, and resource utilization of straw remain pressing challenges in environmental management.

Anaerobic digestion (AD) is a biological process in which organic substances are decomposed by a consortium of microorganisms working synergistically. Briefly, AD consists of four stages: hydrolysis, acidogenesis, acetogenesis, and methanogenesis [3]. AD treatment of straw has been adopted in many countries, and AD technologies for straw are well-developed in developed countries. For example, in Europe, biogas production from energy crops constitutes a significant portion (30–40% wet weight basis of total mixed feed) in agricultural biogas plants [4]. In contrast, the application of AD technology for straw in China is still in the developmental stage.

Although AD is an effective approach to straw treatment, direct AD has limitations regarding the degradation efficiency of raw straw and biogas production

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performance. Low gas yields and prolonged digestion cycles are major challenges associated with direct AD of straw. This is primarily due to the difficulty in degrading lignocellulosic fibres present in the straw and the diminished availability of low molecular weight soluble compounds to anaerobic microorganisms [5].

Current pretreatment methods for lignocellulosic materials such as straw primarily include physical methods, chemical methods, and biological methods. Composting is a form of biological pretreatment that involves the conversion of organic materials such as straw, agricultural by-products, digestate, and animal manure into a more decomposed product. Composting significantly enhances the quality of the final compost products through various decomposition processes [6]. However, there is an ongoing debate regarding straw composting as a method for straw pretreatment. Studies by Yu et al. [7] and Andersen et al. [8] have shown that composting-pretreated straw can increase the production of biogas. Conversely, research by Gao et al. [9], Ajmal et al. [10], and Chen et al. [11] suggests that composting does not significantly improve gas production from straw. Numerous factors influence the efficiency and outcome of composting, including feedstock, temperature, carbon-to-nitrogen (C/N) ratio, and the moisture content. Research indicates that initial moisture content exerts the most significant influence on compost maturation. Maintaining a moisture content of 60% in the compost mixture is optimal for microbial activity [12, 13]. To maintain the moisture content within the optimal range of 50–65% (wet alkaline), water can be added during compost turning, and the containers should be covered with lids to retain moisture and prevent excessive heat loss [14, 15].

The moisture content of straw affects the subsequent pretreatment and fermentation processes. Yuan et al. [16] demonstrated that a 70% moisture content and 4% ammonia concentration in straw led to a significant increase of 28.55% in biogas yield compared to the control treatment. Li et al. [17] investigated the impacts of moisture content and additives on H_2 production, emphasizing that moisture content was the most significant factor. Wang et al. [18], in their study on composting technology in which deer manure was combined with straw, determined an initial moisture content of 55%, an initial C/N ratio of 30:1, and a straw particle size of 1.5–3.5 cm as the optimal operational conditions. Their findings indicated that the initial moisture content had the greatest impact on compost maturity. Currently, there is no report concerning the effects of straw composting pretreatment + extrusion on the anaerobic fermentation to produce biogas.

While both composting and extrusion have shown potential in enhancing the AD of straw individually, their mechanisms and effects differ. Composting offers a biological degradation route, while extrusion provides a rapid physical disruption. Currently, there is a lack of systematic research investigating the sequential application of composting followed by extrusion as a combined pretreatment strategy for wheat straw. The potential synergistic effects of this combination on the physicochemical structure of straw and the ultimate biogas yield remain unexplored.

This study aims to evaluate the effects of different moisture contents on wheat straw, providing critical data for subsequent pretreatment and anaerobic digestion; it further systematically assesses the individual and combined effects of composting treatment and extrusion pretreatment on biogas yield from wheat straw, while investigating their impacts on the physicochemical properties of the straw and anaerobic digestion efficiency, thereby filling this knowledge gap. The specific objectives are: (1) to determine the optimal moisture content for wheat straw prior to extrusion pretreatment; (2) to assess the impact of composting temperature on the composition and methane potential of straw; and (3) to elucidate the synergistic effects of the combined pretreatment on straw structure and its subsequent enhancement of anaerobic digestion performance.

2. Materials and Methods

2.1. Feedstock and Inoculum

The wheat straw (WS) was taken from Funan County, Anhui Province, China. The WS was dry and chopped manually to a size of about 5cm–10 cm, then set aside in a dry and cool place. The total solids (TS), volatile solids (VS) and pH of WS were 93.45 ± 0.58%, 89.03 ± 0.47% and 7.68 ± 0.12, respectively. The inoculum was collected from the AD pool at Anhui Linhai Ecology Co. and transported to the laboratory, where it was mixed with the inoculum in the laboratory for 1 week at 37°C before use. The TS, VS, and pH values of the inoculum were 4.64 ± 0.08%, 3.36 ± 0.01%, and 7.50 ± 0.03, respectively.

2.2. Experimental design

The experiments investigated the effect of WS at different moisture contents on biogas production after extrusion, and also evaluated the use of WS for compost pretreatment and extruding the composted WS for biogas production based on the determined

optimal moisture content. The experimental treatments included the adjustment of the straw moisture content, compost pretreatment, straw extrusion pretreatment, and anaerobic digestion for biogas production.

2.2.1. Adjustment of the straw moisture content

WS moisture content was adjusted in 48 L plastic containers. A given amount of wheat straw and an appropriate amount of water were added to the containers to adjust the moisture content of the wheat straw to the treatment value. The moisture content of the wheat straw was adjusted to 20, 40, 55, 70, and 85%. The containers were covered with lids and left to stand for 24 h. One portion of the WS was then sampled for subsequent extrusion experiments, and the other portion was placed in a plastic bag for the anaerobic digestion experiments.

2.2.2. Compost pretreatment

The composting experiment was carried out at the seedling collection site of Anhui Linhai Ecological Technology Co., Ltd. The moisture content of the wheat straw was measured before composting. Mix 180 kilograms of crushed wheat straw (dry matter mass) with tap water to adjust the moisture content to the optimal level determined by previous experiments. In those experiments, we evaluated the anaerobic digestion performance of wheat straw under various moisture conditions and identified the optimal moisture level that maximizes microbial activity and promotes efficient anaerobic fermentation and methane production. After thorough mixing, the straw was piled into $2 \times 2 \times 1$ m stacks, and the temperature was recorded every hour for the first 10 h, then every 12 h after the stack temperature reached 55°C. Samples (straw in the red plastic net bag at the center of the pile) were taken at the starting point of the reactor temperature, at 30, 40, 50, and 55°C and above for 2 days, with a sample size of 1 plastic box (48 litres). The extracted straw was levelled and cooled before being placed in the plastic box. The pH values of the straw samples were measured in the company laboratory, and samples were stored at −18°C in a freezer. The compost samples were analyzed for TS, VS, total organic carbon (TOC), total nitrogen (TN), cellulose, hemicellulose, and lignin, with small amounts used for FTIR, electron microscopy, and XRD analyses and measurements of specific surface area. Before composting, 10 mesh bags were filled with straw of known dry matter mass with the previously obtained optimal moisture content. The mesh bags were inserted into the middle of the compost piles, and two mesh bags were removed during each sampling period to determine the loss rate of total solids.

2.2.3. Extruding treatment

The extrusion process was conducted at Linhai Ecological Technology Co., Ltd., located in Funan County, Anhui Province, China. The experimental extrusion equipment was described in detail by Chen et al. [19]. A twin-screw extruder with two counter-rotating screws (TSP180, Jinwo Machinery Co., Ltd. Nanjing, China) was used for the experiment. The twin-screw extruder utilizes two parallel, meshed, and co-rotating modular screws and specially structured screw elements to dynamically squeeze the fibre materials while conveying them and to complete fibre crushing and dissociation by kneading between fibres under high concentration and pressure. The power supply voltage of the twin-screw extruder was 380 V, the frequency was 50 Hz, and the maximum power used was 151 kW. The main technical parameters included a screw diameter of 180 mm, a screw speed of 450 r/min, and a screw center distance of 158 mm. Untreated, moisture-adjusted, and compost-pretreated WS samples were continuously fed into the barrel through the inlet orifice by transport screws, and the accumulated WS samples were compressed in the space between the transport screws and the reversed screw element. All extruded WS was collected, placed in airtight plastic bags, and stored at 4°C for the analysis of physicochemical properties and anaerobic digestion experiments. Samples of raw WS and extruded WS were collected and dried at 60°C for 3 days. Part of the sampled straw was used for SEM and SSA determinations, while another part was crushed and passed through a 100-mesh sieve for FTIR and XRD analyses.

2.2.4. Anaerobic digestion

Samples involved in this study included the following categories: untreated straw, straw with different moisture contents, straw with different moisture contents after extruding, samples of straw compost at different temperatures, and samples of straw compost at different temperatures after extruding. All these samples were used for subsequent analyses, including physicochemical property determination and anaerobic digestion experiments.

The anaerobic digestion (AD) experiments were based on existing literature studies [19]. The AD experiments were performed in triplicate using sequencing batch model with a total volume of 1000 mL, while the active volume of the reactor was 800 mL. First, 48 g of wheat straw (based on dry mass), 500 g of inoculum, and distilled water were added to the bioreactor to achieve an initial TS loading rate of 6.0% for WS. The bioreactor was then flushed with nitrogen (99.9%) for 3 min, immediately sealed with a rubber stopper, and

subsequently incubated at (37 ± 1)°C to produce biogas. A gas outlet was installed at the top of the bioreactor, through which the generated biogas was collected into a gas collection cylinder. Meanwhile, two blank bioreactors consisting of inoculum and distilled water were incubated under the same temperature conditions to quantify the background biogas production, and the values were subtracted from the total gas production. The experiment lasted for 41 days and was terminated when no significant biogas production was observed for 1 consecutive week. To prevent material stratification and excessive floating, each reactor was manually agitated twice daily.

The wheat straw with moisture contents of 30, 40, 55, 70, and 85%, as well as the above wheat straw after extrusion, are referred to as pre- and post-extrusion wheat straw, comprising the natural state of straw and the extruded straw. The anaerobic digestion experiments were conducted by inoculating each straw treatment separately, and the biogas production data were collected for analysis. The uncomposted, composted, and extruded compost samples were inoculated for the anaerobic digestion experiments. The wheat straws that reached ambient temperatures, 30, 40, 50, and 55°C and above after 2 days were labelled as T0, T1, T2, T3, T4, and T5. The wheat straw that reached ambient temperature, 30, 40, 50, and 55°C and above after 2 days was labelled as L0, L1, L2, L3, L4, and L5 after being extruded and composted. Samples were collected on days 0, 2, 5, 9, 16, 25, and 41 of the anaerobic digestion experiment using compost extruded materials, and the pH, electrical conductivity (EC), total volatile fatty acids (TVFAs), soluble chemical oxygen demand (SCOD), and alkalinity of the fermentation broth were determined. After the experiment, the solid samples were filtered through a coarse cotton cloth, and the TS, VS, and ash content were measured and used to calculate the total solid (TS) loss of the fermented straw. After drying and crushing a portion of the solid sample through a 100-mesh screen, the sample was used for Fourier transform infrared spectroscopy (FTIR), X-ray diffraction (XRD), and scanning electron microscopy (SEM) analyses.

2.3. Analytical methods

A mercury thermometer was placed at the center part of the composting pile to monitor the core temperature, and three additional mercury thermometers were used to measure the temperature at three different central points of the pile. The determination of total solids (TS) was carried out by drying the sample at 105°C for 24 h to a constant weight using the differential weight method. The volatile solids (VS) were determined by igniting the dried sample at 550°C for 4 h in a muffle furnace, and the difference in weight before and after ignition was calculated as the VS. The pH value of the dry WS samples was determined by preparing an aqueous slurry. Approximately 10 g of the dry WS sample was mixed with 100 mL of distilled water, and the mixture was stirred for 30 min. The pH of the slurry was then measured using a pH metre (PHB-4, Leici, China). Daily biogas production was measured by water displacement and recorded daily. The cumulative biogas production was calculated as the sum of the daily biogas production over the entire experimental period. A continuous gas collection method is employed when biogas production varies across different stages. For instance, during phases of lower biogas production, biogas is collected continuously over several days, with methane concentration measurements conducted uniformly on the final collection day. The collected gas was passed into a GC-5890N gas chromatograph for measurement. The column used in the gas chromatograph was FFAP, and the parameters for measuring methane concentration were an inlet B temperature of 105°C, a detector B temperature of 120°C, b uo a column box temperature of 80°C, and a single sample measurement time of 5 min (GC5890N, Nanjing Kejie Analytical Instrument Co., Ltd, China). The calculation formula for methane concentration is as follows:

where the methane content; b is the methane concentration on that day, %; a is the daily biogas concentration, mL/d, and M is total biogas production, mL. The square-root term of daily biogas production ($\sqrt{a}$) is an empirical correction factor adopted in the calculation method used in this study. It was introduced to account for the nonlinear relationship between daily biogas production and methane yield under the experimental conditions. This equation was applied consistently for all treatments to ensure the comparability of methane estimation.

The determination of TVFAs was conducted following the method described in a previous study [20]. All samples were measured in triplicate. The specific surface area (SSA) was determined using an automatic Brunauer–Emmett–Teller specific surface area and porosity analyzer (ASAP 2460, Micromeritics, USA) via the nitrogen adsorption at 77 K.

Wheat straw was air-dried, crushed, and passed through a 100-mesh sieve, then used for FTIR (NEXUS 870FT, Nicolet, USA) and X-ray diffraction (XTRA, Thermo Fisher, USA) analysis. Crystallinity was calculated following the method described in a previous study [19]. The surface structure of wheat straw was observed using a scanning electron microscope (Tescan Mira4, Tescan,

Czech Republic). The surface morphology of treated wheat straw was observed using a 2 μm scanning electron microscope (SEM) (S3400, Hitachi, Japan).

2.4. Mapping and data analysis

One-way analysis of variance (ANOVA) ($P < 0.05$) and factor analysis were performed on data from three replicates of each treatment; results were reported as mean values ± standard deviation. ANOVA and factor analysis were carried out with the software package SPSS (Version 21, IBM Inc., 2012). Jade 6.0 was used for XRD pattern fitting. Origin 2022 was used for plotting of experimental data, and Excel 2019 software was used for processing of experimental data.

3. Results and discussion

3.1. Effects of wheat straw with different moisture contents on AD before and after extrusion

Figure 1 shows biogas production from wheat straw at various moisture contents, before and after extrusion. Extrusion increased biogas production. This improvement is primarily attributed to the reduced particle size of the wheat straw after extrusion the treatment altered the aggregation state and porosity of the wheat straw matrix, promoting an increase in surface area and a decrease in crystallinity. This finding is consistent with the results reported in a previous study [21]. Minimal changes in biogas production were observed at 40% and 85% moisture content, indicating

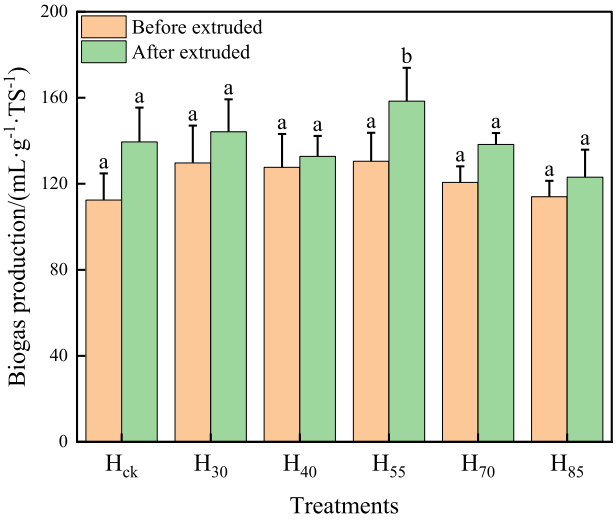


Figure 1. Changes in biogas production before and after rubbing with different moisture contents. The letters indicate significant differences among feed parameters with significance level $P < 0.05$.

that both extremely high and low moisture levels limit biogas production. Excessive moisture also increases extrusion power consumption [18]. The moisture content of composting materials also affects the bioavailability of oxygen in microbial processes.

Water is essential for the decomposition process, and moisture stress is one of the most common limiting factors for microbial activity on solid organic substrates. However, when the moisture content exceeds 65%, air in the pore spaces of the raw material is displaced by water, resulting in anaerobic conditions, odour issues, and reduced decomposition rates [22, 23]. Maintaining a moisture content of approximately 60% in the substrate mixture is optimal for microbial activity [6, 12]. Among the five treatments, 55% moisture content in wheat straw showed the highest biogas production after extrusion (158.41 $mL \cdot g^{-1} \cdot TS^{-1}$), a 21.4% improvement compared to pre-extrusion. Thus, the pre-treatment can enhance biogas production from wheat straw, and different moisture contents of the wheat straw also exert a significant influence on biogas production. The results indicated that wheat straw with a moisture content of 55% and extrusion pre-treatment exhibited optimal biogas production, providing supporting data for subsequent experiments.

3.2. Effects of compost pretreatment on straw

3.2.1. Effect on dry matter loss

The variation in compost pile temperature during the composting process is presented in Figure 2a. During the first 10 h of the experiment, the compost pile temperature increased rapidly, and reached 55°C at 21 h. By 29 h, the temperature of the compost pile inside the reactor had reached 60°C. At this point, the reactor was turned over, leading to a temperature drop to 20°C. The purpose of turning over the compost pile was to alter the positioning of internal and external wheat straw, thereby enhancing biogas exchange between the compost pile and its surroundings, improving microbial growth conditions, and increasing the degradation rate of organic matter [8]. After turning, the compost pile temperature increased rapidly again, stabilizing at approximately 57°C and remaining stable for 40 h.

According to the experimental design, samples were collected at the initiation of the experiment (T0), as well as when the pile temperature reached 30°C (T1), 40°C (T2), 50°C (T3), 55°C (T4), and when it exceeded 55°C for 2 days (T5). Corresponding composting experiments were conducted at 0, 11, 14, 18, 21, and 69 h, with the remainder used for anaerobic digestion and biogas production experiments.

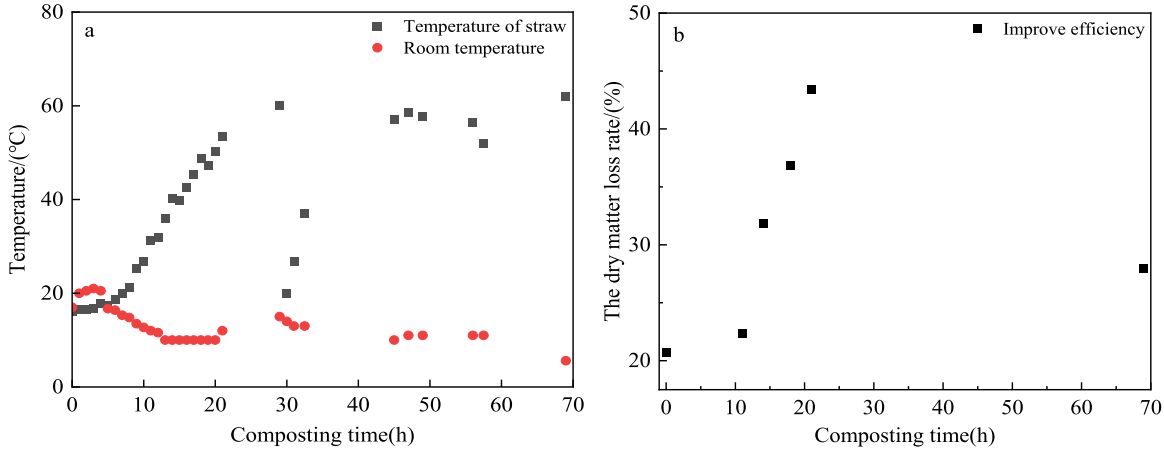


Figure 2. Changes in compost temperature (a) and dry matter loss rate of wheat straw (b) during the composting process.

The results of TS loss during WS composting are presented in Figure 2b. The composting process involves the microbial decomposition and utilization of WS organic matter, as well as heating of the WS pile. A substantial loss of WS dry matter occurred during composting, primarily due to the decomposition of organic matter. WS's TS loss rate exhibited a gradual increase during the initial 21 h of composting, and reached 28% by 69 h. The TS loss rate continued to increase with higher pile temperature and longer duration. Higher pile temperatures accelerated the decomposition of organic matter, thereby resulting in a further significant loss of WS dry matter [24]. This study primarily focused on the influence of composting pre-treatment on AD, where dry matter loss inevitably leads to reduced methane production from WS anaerobic digestion. Therefore, no additional samples were collected after 69 h of composting.

3.2.2. Effect of composting pre-treatment on wheat straw AD

The biogas production trends of WS under different treatments displayed analogous patterns, with initial rapid increases during the early fermentation phase (Figure 3a). Days 6–10 corresponded to the peak biogas production phase, followed by a sharp decline in methane production. By day 25, the daily biogas production dropped to below 50 mL and stabilized. Figure 3b illustrates the total biogas production from raw and composted WS under different composting temperatures. All composted WS treatment groups exhibited higher methane yields in comparison to raw WS, with the group composted at 55°C achieving the highest production of 128.00 mL/g. This represented a 21.74% relative increase in total biogas production in comparison to raw straw. The results demonstrate that composting pre-treatment significantly enhances biogas production from wheat straw AD, which is consistent with findings reported by Yu et al. [7] and Andersen et al. [8].

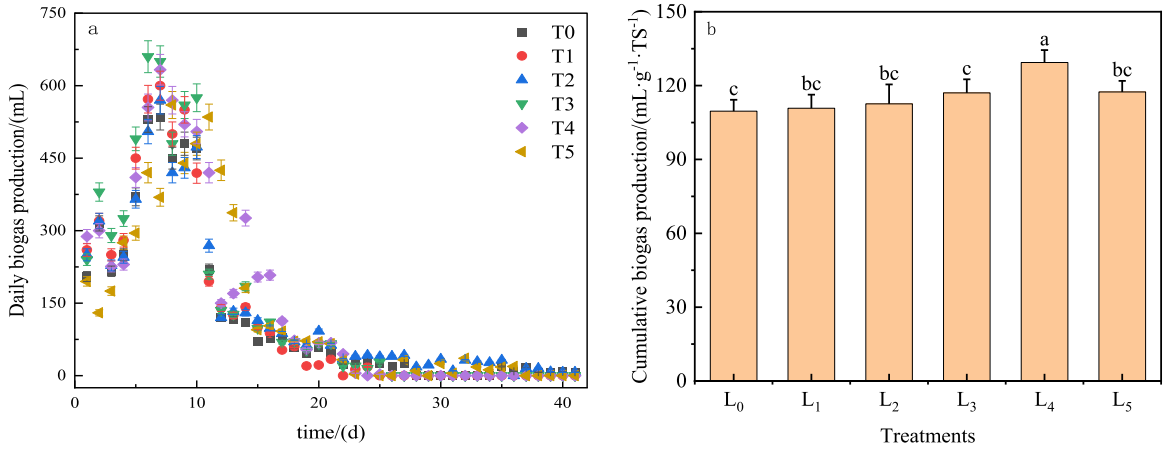


Figure 3. Changes of daily (a) and cumulative (b) biogas yield after composting process.

Table 1. Comparison of biogas production rate improvement of different pre-treatment.

Treatments	Biogas production after composting pre-treatment /mL	Improvement of biogas production by compost pre-treatment	Biogas production after combined pre-treatment /mL	The improvement of biogas production by extrusion pretreatment	The improvement from combining two pre-processing methods
0	5047		6059	20.76%	20.76%
30	5211	3.25%	6177	18.54%	22.39%
40	5538	9.73%	6654	20.15%	31.84%
50	5921	17.32%	6906	16.64%	36.83%
55	6614	21.74%	7240	17.84%	43.45%
55-2d	5598	10.92%	6460	15.40%	28.00%

3.3. Effect of extrusion pre-treatment on the AD of wheat straw

3.3.1. Biogas production

Samples from group T0 were collected before pile formation and heating. At this stage, the wheat straw served as the initial substrate for AD experiments following mixing pretreatment, and biogas production was measured.

The results demonstrated a total biogas production of 5220.67 mL from the initial WS, and this value increased to 6438.33 mL after mixing pretreatment, marking a 23.32% improvement in biogas production (Table 1). These results served as a reference for subsequent blank groups. Biogas production from WS subjected to different composting temperatures was compared before and after mixing pretreatment. The percentage improvement in biogas production was detailed in Table 1. Apart from the blank group data, mixing pretreatment enhanced the AD performance of WS, with the improvement in biogas production ranging from 18% to 39%. The differences in improvement between the blank group and other treatment groups were minor but consistent. Thus, mixing pretreatment effectively enhances biogas production from the anaerobic digestion of WS.

3.3.2. pH values

Research indicates that pH directly regulates the process and intermediate products of AD. The optimal pH range for this process has been reported to range from 6.8 to 7.4 [25]. Figure S1 depicts the pH variation in the AD system using compost-pretreated and untreated WS before mixing. The pH trends among different treatment groups exhibited an initial convergence, with the pH value first decreasing, then increasing, and finally stabilizing. The pH value decreased continuously from day 0 to day 5, reaching its minimum on day 5. From day 6 to day 9, the pH value increased and stabilized thereafter. The post-mixing pH variation exhibited a similar trend (Figure S2). This phenomenon was primarily attributed to the substantial production of organic acids during the initial stage of AD, thereby leading to a pH decrease in the early stage of the AD reactor. However, as the process continued, the production of ammonia increased, raising the pH value to 7.0–7.2. Methane production stabilizes when the pH reaches 7.2–8.2. Post-mixing pH fluctuations exhibited smaller magnitudes and greater stability compared to pre-mixing conditions. This observation might be attributed to the disruption of the WS surface rigid structures after mixing, which facilitated the breakdown of WS by fibre-degrading microbial communities, thereby lowering the pH value.

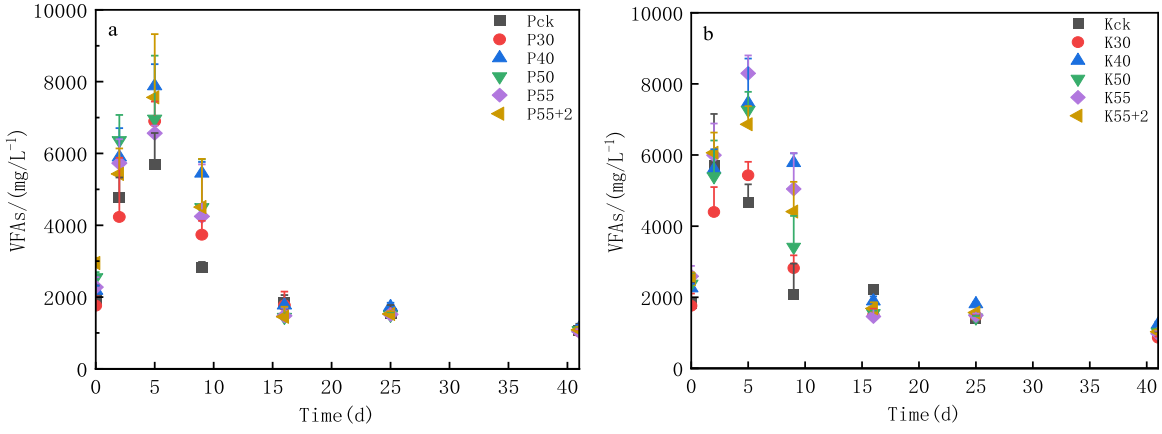


Figure 4. Changes of VFAs contents obtained before extruding (a) and after extruding (b).

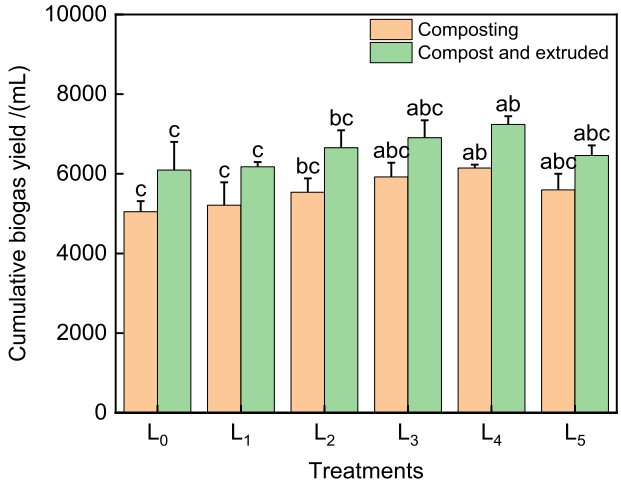


Figure 5. Change of biogas production improvement rate of combined pre-treatment.

3.3.3. VFAs variation

The variation in VFAs during the AD process before and after mixing is presented in Figure 4(a) and (b). The trends in VFA contents for WS were similar before and after mixing, exhibiting an initial increase followed by a decrease and subsequent stabilization. During fermentation, the VFA content increased until the 5th day, with the highest value observed in wheat straw subjected to mixing pretreatment at a composting temperature of 50°C, which reached 8296.67 mg/L. From day 6 to day 16, the VFA content decreased gradually and stabilized after day 17. However, the magnitude of variation in VFA content after mixing was smaller in comparison to that before mixing. This could be attributed to the transformation of WS into a softer fibrous material after mixing, as the process separates cellulose, hemicellulose, and lignin, thereby facilitating more efficient decomposition of organic matter by microbes and promoting VFA accumulation.

Under controlled fermentation conditions, VFAs accumulated gradually in the substrate during the initial stage of fermentation, primarily dominated by acetic acid throughout the fermentation process. This was attributed to the rapid growth of acidogenic bacteria during the biogas fermentation process, which convert substrates into VFAs. The accumulation of VFAs occurred because the accumulated VFAs could not be timely converted into methane.

Table 2. Comparison of specific surface area.

	Specific surface area			
Group	Single point surface area at *P*/Po = 0.314596977	BET surface area	Langmuir surface area	t-plot external surface area
Control group	4.8101	4.603	21.8306	7.4582
L0	3.1904	2.6983	12.0704	3.9752
T4	3.067	2.6133	7.852	3.5456
L4	8.7803	9.9347	83.3355	16.2127

3.4. Effect of composting and extrusion on AD of wheat straw

3.4.1. Effect on biogas production

In this study, AD was conducted on WS in the L0, L1, L2, L3, L4, and L5 groups, and the biogas production rates of WS in these groups were compared. Composting combined with extrusion pretreatment enhanced the biogas production from wheat straw. Composting treatment enhanced the biogas production of WS at room temperature, 30, 40, 50, and 55°C and above for 2 days (Figure 5). The biogas production initially increased with increasing composting temperature, attaining a maximum value of 6144 mL when the temperature reached 55°C. However, when temperatures above 55°C were maintained for an additional 2 days, the production of biogas decreased to 5598 mL. After composting pre-treatment combined with extrusion pretreatment, the biogas production at various composting temperatures increased significantly, exhibiting a trend of first increasing and then decreasing. After extrusion pretreatment at 55°C, the biogas production of WS was 7240 mL, which was 43.45% higher than that of the control group. When composting temperatures were further increased and maintained for 2 days, combined with extruding pretreatment, the biogas production of WS decreased to 6460 mL.

To investigate the reasons behind the enhanced AD biogas production due to pretreatment, specific surface area tests were conducted on WS samples from each group. The results (Table 2) indicated that composting pretreatment alone did not increase the specific surface area of WS. The BET specific surface areas of L0 and T4 were 2.6983 and 2.6133 m^2/g, respectively. However, when combined with extrusion pretreatment, there was a significant increase in the specific surface area of WS, consistent with the experimental results of biogas production. The BET specific surface area of L4 was 9.9347 m^2/g, which indicated a highly significant increase. Therefore, one reason for the enhanced biogas production from WS treated with combined composting and extrusion pretreatment is the increase in specific surface area.

3.4.2. FTIR analysis

The results of FTIR analysis of WS samples under various treatments (original WS, composted at 55°C, composted at 55°C with extrusion, and composted at 55°C with extrusion and anaerobic digestion (abbreviated as after

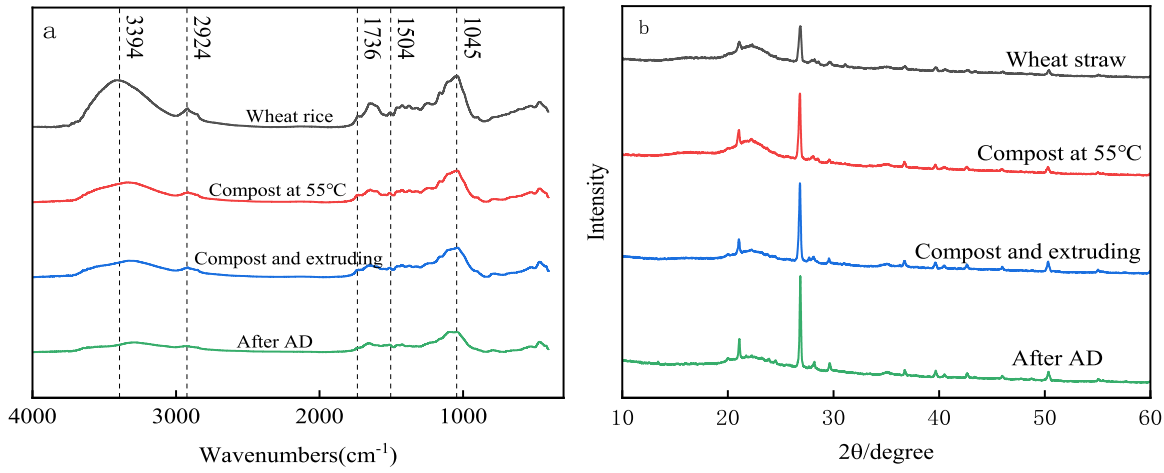


Figure 6. The FTIR (a) and X-ray (b) diffraction of wheat straw with different treatments.

AD)) are presented in Figure 6a. The overall spectral features of WS under different pretreatments were similar, although the absorbance and/or wavenumbers of certain characteristic absorption bands exhibited changes. The peak intensities of the original WS, composted at 55°C, composted at 55°C with extrusion, and after AD at 3394, 2924, 1376, and 1045 cm^{-1} exhibited a gradual decreasing trend, with the highest intensity observed for original WS and the lowest for the AD group. All WS samples exhibited a broad absorption band at around 3394 cm^{-1}, which is attributed to the O–H stretching vibrations of hydrogen-bonded hydroxyl groups, primarily from cellulose. This absorption band shifted slightly to a lower wavenumber value of 3394 cm^{-1}. The absorption band near 2924 cm^{-1} represents the C–H stretching vibrations of cellulose [26]. The absorption band around 1736cm^{-1} is assigned to the characteristic bending or stretching vibrations of functional groups in cellulose [27]. The absorbance intensity of hemicellulose at 1045 cm^{-1} was significantly higher for original WS than that for composted at 55°C, composted at 55°C with extrusion, and after AD samples. The absorbance intensity of hemicellulose at 1045 cm^{-1} in the original WS was close to that for the WS treated with compost at 55°C with extrusion and after AD, although a distinct diagnostic peak for hemicellulose was observed at this wavenumber.

The infrared absorption band near 1504 cm^{-1} is assigned to the C = C stretching vibration of aromatic rings in lignin [28]. The band near 1240 cm^{-1} is a diagnostic peak for the C–O–C stretching vibration of cellulose [29]. Hemicellulose, a polysaccharide with a complex FTIR absorption pattern, exhibits characteristic absorption peaks in the range of 1200–1900 cm^{-1}. This region primarily involves ring vibrations and overlaps with the stretching vibrations of C–OH side groups and C–O–C glycosidic bond vibrations. The absorption peak at 1045 cm^{-1} is closely associated with hemicellulose [30].

The infrared absorption band intensities of original WS were greater than those of the WS composted at 55°C, composted at 55°C with extrusion, and after AD. This indicates a decrease in the hydrogen bonding of cellulose and the cleavage of methyl or methylene groups, which may contribute to a decrease in WS crystallinity. Additionally, the infrared absorption band intensity at 1504 cm^{-1} was significantly decreased, suggesting the cleavage of important ester bonds in lignin.

3.4.3. XRD analysis

Pretreatment modifies the crystalline structure and crystallinity of cellulose, thereby affecting cellulose's enzymatic hydrolysis efficiency. Therefore, X-ray diffraction (XRD) analysis was employed to assess the crystallinity and structure of WS before and after pretreatment. Wheat straw samples subjected to composting at 55°C, composting at 55°C with extrusion, and after AD exhibited typical cellulose structures (Figure 6b). In the XRD patterns, WS samples exhibited a prominent peak at 26.8° and a minor peak at 22.5°. The positions of these peaks did not differ significantly among the various treatments, indicating that the crystalline regions of WS were not significantly destroyed during the pretreatment and AD processes. Compared to untreated WS, minimal changes were observed in crystallinity, peak intensity, and grain size following pretreatment and subsequent AD fermentation. Although crystallinity, peak intensity, and grain size all increased with WS pretreatment. The crystal plane peaks of WS composted at 55°C, composted at 55°C with agitation, composted at 55°C with extrusion, and after AD were narrower and

sharper than those of untreated WS, with significantly enhanced crystal plane intensity, indicating an increased proportion of cellulose crystallinity. This phenomenon is likely attributed to the partial removal of hemicellulose and lignin. Pretreatment and subsequent AD fermentation increased cellulose exposure and enzyme accessibility, thereby promoting enzymatic hydrolysis efficiency.

Each treated WS sample exhibited a strong peak at approximately 2θ = 26.60°. Comparison with the PDF – # 99-0088 standard colour chart confirms that this was the diffraction intensity peak of SiO_2. The peak intensities of each treatment at this location were in the order of after AD > Composite at 55°C with extrusion > Composite at 55°C > WS, indicating that Si was released from WS after the experiment. In addition, a peak appeared in each treated WS sample at approximately 21.0°, and the order of peak size was after AD > composite at 55°C with extrusion > composite at 55° C > WS. Comparing PDF card # 03-0226 confirms that this was the (101) crystal plane diffraction of cellulose I. Anaerobic digestion treatment intensified this process, which is consistent with the FTIR results.

4. Conclusion

This study confirmed extrusion pretreatment enhanced the biogas production of WS through anaerobic digestion, with the best effect achieved at an initial moisture content of 55%, increasing the gas production by 21.4% compared to the non-extruded group. Composting enhanced WS gas production via structural destruction, with 55°C as the advisable termination temperature, increasing output by 21.74%. Extrusion-composting combined pretreatment had a synergistic effect. Under the condition of being maintained at 55°C for 21 h, the gas production of WS was significantly increased by 43.45% compared to the control group, which was superior to the single pretreatment methods. The sequential combined pretreatment of composting and extrusion is a high efficiency pretreatment strategy of straw by synergistic biological and physical action.

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Author contributions

CRediT: **Ya-Ge Li:** Formal analysis, Investigation, Writing – review & editing; **Guang-Cai Ge:** Data curation, Formal analysis, Validation; **Yu-Juan Sun:** Data curation, Formal analysis; **Yu-Xin Zhang:** Formal analysis, Investigation, Validation.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Data availability statement

All authors are willing to provide the results presented in the paper or the data and materials analyzed in the paper free of charge upon request.

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